KYLE DAWSON, PHD

PhD Scientist & Cloud Solution Builder | Satellite Remote Sensing, Aerospace & Machine Learning

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team, setting technical direction and architecture standards for satellite-, LiDAR-, and imagery-driven systems
- architected and deployed cloud-native, containerized services (Cloud Run, Docker, CI/CD, BigQuery) that ingest and process large-scale geospatial and remote-sensing data through scalable, reliable pipelines
- built a GPU-accelerated 3D computer-vision pipeline that maps infrastructure by aligning engineering CAD models to LiDAR point clouds
- prototyped and deployed a multi-model generative-AI application integrating managed foundation-model services (Vertex AI, Cohere) with a retrieval benchmark for document Q&A and image analysis

Data Scientist → Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- designed, deployed, and delivered customer-facing cloud analytics platforms (Plotly Dash on Azure) that turned complex satellite- and drone-based sensing capabilities into productized solutions, directly securing contracts of $50–150k
- served as customer-facing technical lead across 10+ countries, presenting solutions, architecture, and technical strategy to executives, enterprise customers, and international conference audiences
- executed proofs of concept and authored technical documentation and uncertainty assessments supporting multi-year enterprise contracts

Data Scientist — Ag Tech

Bayer CropScience

Jan 2020 – Mar 2022 St. Louis, MO

- built end-to-end cloud data workflows on AWS (S3) combining relational databases, geospatial data, and aerial imagery into analytics-ready datasets
- applied machine learning and computer vision (Detectron2) to UAV- and aircraft-derived imagery for large-scale digital phenotyping
- delivered interactive decision-support applications to internal stakeholders and business teams

Postdoctoral Fellow — Aerospace & Remote Sensing

NASA Langley Research Center

Feb 2018 – Jan 2020 Hampton, VA

- developed retrieval algorithms for airborne and satellite remote-sensing instruments (combined High Spectral Resolution and Raman LiDAR) supporting NASA Earth-observation and atmospheric-science missions
- delivered peer-reviewed results and instrument data tooling for the aerospace remote-sensing research community
- prior NASA appointments at Langley (LARSS, HSRL) and Ames (active/passive remote-sensing algorithm comparison), spanning three NASA research centers

SELECTED PROJECT

NASA HSRL Data Toolkit (hsrltool)

Open-source scientific software

A graphical toolkit for reading, inspecting, and visualizing NASA High Spectral Resolution Lidar (HSRL) instrument data (HDF5), used by the atmospheric remote-sensing community.

EXECUTIVE SUMMARY

PhD scientist and technical leader with 8+ years architecting, building, and delivering cloud-based data, ML, and remote-sensing systems across AWS, Azure, and GCP — paired with deep aerospace and satellite remote-sensing domain expertise from NASA. Skilled at translating complex technical capabilities into customer-facing cloud solutions: designing production platforms, executing proofs of concept, securing and delivering enterprise contracts, and presenting architecture and business value to executives, customers, and partners worldwide.

CORE COMPETENCIES

Cloud Solution Architecture
Customer-Facing Solution Delivery
Proofs of Concept
Technical Strategy & Enablement
Reference Architecture & Documentation
Technical Leadership
Cross-Functional Collaboration
Executive & Partner Communication
Mentoring & Training

DOMAIN EXPERTISE

Satellite & Airborne Remote Sensing
Aerospace / Earth Observation
LiDAR & Point Clouds
Geospatial Analytics
Cloud & Distributed Computing
Systems & Data Integration
Machine Learning
Computer Vision
Generative AI & Foundation Models

TOOLS & TECHNOLOGIES

AWS
AWS S3
Azure
GCP
Cloud Run
BigQuery
Vertex AI
Docker
CI/CD
Python
PyTorch
SQL
GPU / CUDA
GDAL
QGIS
Git
Plotly
Linux

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 Hampton, VA

Click for Roster

Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 Hampton, VA

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing